

1. Product Brief — Mission & Success Criteria

Mission

Provide a multi-application RAG web platform where each “Application” is an isolated knowledge base with:

- Unique Application Name
- Application description
- Starter Questions (4)
- Configuration Instructions (Markdown, stored as flat file)
- Configuration Settings (JSON, validated by JSON Schema)
- Documents uploaded into each Application’s vector store
- Query functions scoped **strictly to the Application**

Success Metrics

- Application resolution by name is **100% unique**
 - Application creation success $\geq 99\%$
 - External applications resolve `/api/apps/by-name/{name}` with zero ambiguity
 - Zero cross-application retrieval leakage (strict `app_id` scoping)
 - Successful ingestion latency $p95 \leq 3s$ (excluding file upload time)
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2. Applications & Scenarios

Key personas

- Internal Admin / Content Curator
- Knowledge Worker
- External Integrator Service (read-only regarding Application Name)

Key Scenarios

Scenario Description

A Create Application with Name, Description, Starter Questions

Scenario Description

- B Modify Application Name (internal-only)
 - C Edit Configuration Instructions (Markdown → flat file)
 - D Edit Configuration Settings (schema-driven JSON form)
 - E Upload documents to Application
 - F Query Application knowledge base
 - G External app resolves Application by name & reads instructions
-

3. Information Architecture / Routes

Application-level pages

/apps

/apps/new

/apps/:app_id

/apps/:app_id/config?tab=instructions

/apps/:app_id/config?tab=settings

/apps/:app_id/upload

/apps/:app_id/docs

/apps/:app_id/docs/:doc_id

/apps/:app_id/search

API endpoints (v3.3 additions included)

GET /api/apps

GET /api/apps/by-name/{name}

POST /api/apps

GET /api/apps/{app_id}

PATCH /api/apps/{app_id} -- rename, modify description, starter questions

GET /api/apps/{app_id}/instructions

PATCH /api/apps/{app_id}/instructions

GET /api/apps/{app_id}/settings

PATCH /api/apps/{app_id}/settings

POST /api/apps/{app_id}/uploads

GET /api/apps/{app_id}/docs

GET /api/apps/{app_id}/docs/{doc_id}

DELETE /api/apps/{app_id}/docs/{doc_id}

POST /api/apps/{app_id}/docs/{doc_id}/reingest

POST /api/apps/{app_id}/search

4. UI Components & States

Applications List

- Displays **Application Name**, description, #docs, updated time
- Does **not** display app_id

New/Edit Application

- Name (unique; external read-only)
- Description
- Starter Questions [4]
- Tabs:
 - **Instructions (Markdown)**
 - Markdown editor + preview
 - Save Instructions → writes to file & updates DB pointer
 - **Settings (schema-driven)**

- Auto-generated form
- Save Settings → JSON schema validation

Upload Page

- Dropzone, queue, ingestion status
- Status chips: uploading, ingesting, ready, error

Search Page

- SearchBar, top_k control
- Starter Questions quick-fill
- Results list with citations
- Debug trace panel

5. Data Model

Application

```
{
  "id": "UUID",

  "name": "string",           // unique, external key, rename allowed internally only
  "slug": "string",          // optional, machine-friendly handle
  "description": "string",

  "starter_questions": ["q1", "q2", "q3", "q4"],

  "instructions_uri": "string", // path to markdown flat file
  "instructions_version": 1,
  "instructions_updated_at": "...",

  "config_settings": { ... }, // validated JSON
  "config_schema": { ... },   // JSON Schema
  "config_instructions_md": null, // optional; deprecated, replaced by flat file
}
```

```
"created_at": "...",  
"updated_at": "..."  
}
```

Document

```
{  
  "id": "UUID",  
  "app_id": "UUID",  
  "app_name": "string",  
  "filename": "string",  
  "mime_type": "string",  
  "size_bytes": 12345,  
  "language": "zh/en/auto",  
  "tags": [],  
  "status": "pending/uploading/ingesting/ready/error",  
  "error_message": null,  
  "created_at": "...",  
  "updated_at": "..."  
}
```

IngestionJob, QueryLog

Same as v3.1, with strict app_id required.

6. External Application Lookup Rules

Allowed

External app:

- uses Application Name to resolve Application
- calls `/api/apps/by-name/{name}`

- receives app_id + metadata + instructions_uri

Not allowed

- External app **cannot modify Application Name**
- External app cannot modify instructions or settings
- External app cannot directly ingest documents

Result

Because rename is internal-only:

Application Name is a stable, unique external lookup key.

7. Configuration Instructions (Flat File Architecture)

Storage:

instructions/{app_id}/instructions.md

DB stores:

instructions_uri

instructions_version

instructions_updated_at

Workflow:

1. PATCH /api/apps/{app_id}/instructions
2. Backend writes file to storage (local or S3)
3. Updates DB pointer + version + timestamp
4. Returns success

External app can:

- read file directly from instructions_uri, **or**
 - call /api/apps/by-name/{name}/instructions
-

8. Configuration Settings (JSON Schema-driven)

Stored in DB:

config_settings JSONB

config_schema JSONB

Edited via:

/apps/:app_id/config?tab=settings

Validation:

- Client-side schema validation
- Server-side schema validation

Used in rag_subsystem calls:

process_files(..., config=application.config_settings, ...)

retrieve_data(..., config=application.config_settings, ...)

9. Ingestion & Retrieval Integration (rag_subsystem)

Ingestion

```
process_files(  
    documents,  
    config=application.config_settings,  
    store,  
    embed_client,  
    router  
)
```

Retrieval

```
retrieve_data(  
    query_text,  
    top_k,  
    filters={"app_id": app_id, ...user filters},  
    config=application.config_settings,  
    store,
```

```
    embed_client,  
    router  
)
```

! Never send instructions_md or instructions_uri to rag_subsystem.

10. Workflow Summary

application.create_v1

- validate name uniqueness
- create app record
- initialize instructions file
- initialize config from schema

application.rename_v1

- allowed only by internal UI
- external apps cannot call this route
- enforce uniqueness

instructions.update_v1

- write markdown to file storage
- update DB pointer

settings.update_v1

- validate JSON via schema
- update DB

ingest_documents_v3

- scope by app_id
- call rag_subsystem.process_files

retrieve_query_v3

- force filters.app_id
- call rag_subsystem.retrieve_data

11. Workflow Execution Mapping v3.3 (Delta)

Only showing new/updated areas:

workflow_id: rag_app_flows_v3_3

version: 3.3

states:

- id: app_lookup_by_name

- type: task

- action:

- type: db_query

- query: "select * from applications where name_lower = :name"

- on:

- not_found: app_lookup_failed

- success: next_state

- id: update_instructions

- type: task

- action:

- type: file_write

- path_ref: "instructions/{app_id}/instructions.md"

- on:

- success: update_instructions_db

- failure: instructions_write_failed

(This extends the full workflow map introduced earlier.)

12. Acceptance Criteria

Application Name Uniqueness

- Given two apps with different names → both can be created
- Given existing “LegalKB” → creating another “LegalKB” returns 409

External Lookup

- Given /api/apps/by-name/LegalKB → returns exactly one application
- Given name not found → 404

Rename Safety

- Given internal rename “LegalKB” → “CorporateLegalKB” → external lookup by new name succeeds
- external app cannot call rename endpoint

Instructions File Export

- Given PATCH instructions → file updated in storage and DB pointer updated
- External app reading file via instructions_uri gets new content

Cross-application isolation

- Searching in App A never returns documents from App B
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13. Security

- Application Name lookup is read-only
 - External apps cannot modify anything
 - instructions_uri access control must be configured (signed URL or ACL)
 - No rag_subsystem modification
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14. Appendix

- All design tokens as in /mnt/data/Design_Tokens.pdf
- Backend patterns from /mnt/data/Backend_Examples.pdf
- Workflow format from /mnt/data/Workflow_Execution_Mapping_Template.pdf
- Runtime prompt template from /mnt/data/AI_Agent_RunTime_Prompt_Template.pdf